CN SDN PROJECT

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Screenshot 1:

```
akash-bagoji@akash-bagoji-HP-Pavilion-Laptop-15-eg3xxx:~/pox$ cd ~/pox
python3 pox.py forwarding.l2_learning
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
```

Screenshot 2:

```
2.3.0
akash-bagoji@akash-bagoji-HP-Pavilion-Laptop-15-eg3xxx:~$ sudo mn --topo single,3 --mac --controller remote
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
Interrupt
mininet>
Interrupt
mininet>
```

Screenshot 3:

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

Screenshot 4:

```
akash-bagoji@akash-bagoji-HP-Pavilion-Laptop-15-eg3xxx:~$ sudo ovs-ofctl add-flow s1 "dl_dst=ff:ff:ff:ff:ff:ff,actions=drop"
akash-bagoji@akash-bagoji-HP-Pavilion-Laptop-15-eg3xxx:~$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=6.893s, table=0, n_packets=0, n_bytes=0, dl_dst=ff:ff:ff:ff:ff:ff actions=drop
```

Screenshot 5:

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

Screenshot 6:

```
akash-bagoji@akash-bagoji-HP-Pavilion-Laptop-15-eg3xxx:~$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=115.975s, table=0, n_packets=0, n_bytes=0, dl_dst=ff:ff:ff:ff:ff:ff actions=drop
```

Screenshot 7:

```
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 iperf -s &
mininet>
mininet> h1 iperf -s &
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
bash: hiperf: command not found
mininet> h2 iperf -c h1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 59954 connected with 10.0.0.1 port 5001 (icwnd/mss/irtt=14/1448/4629)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0127 sec  92.2 GBytes  79.1 Gbits/sec
mininet>
```